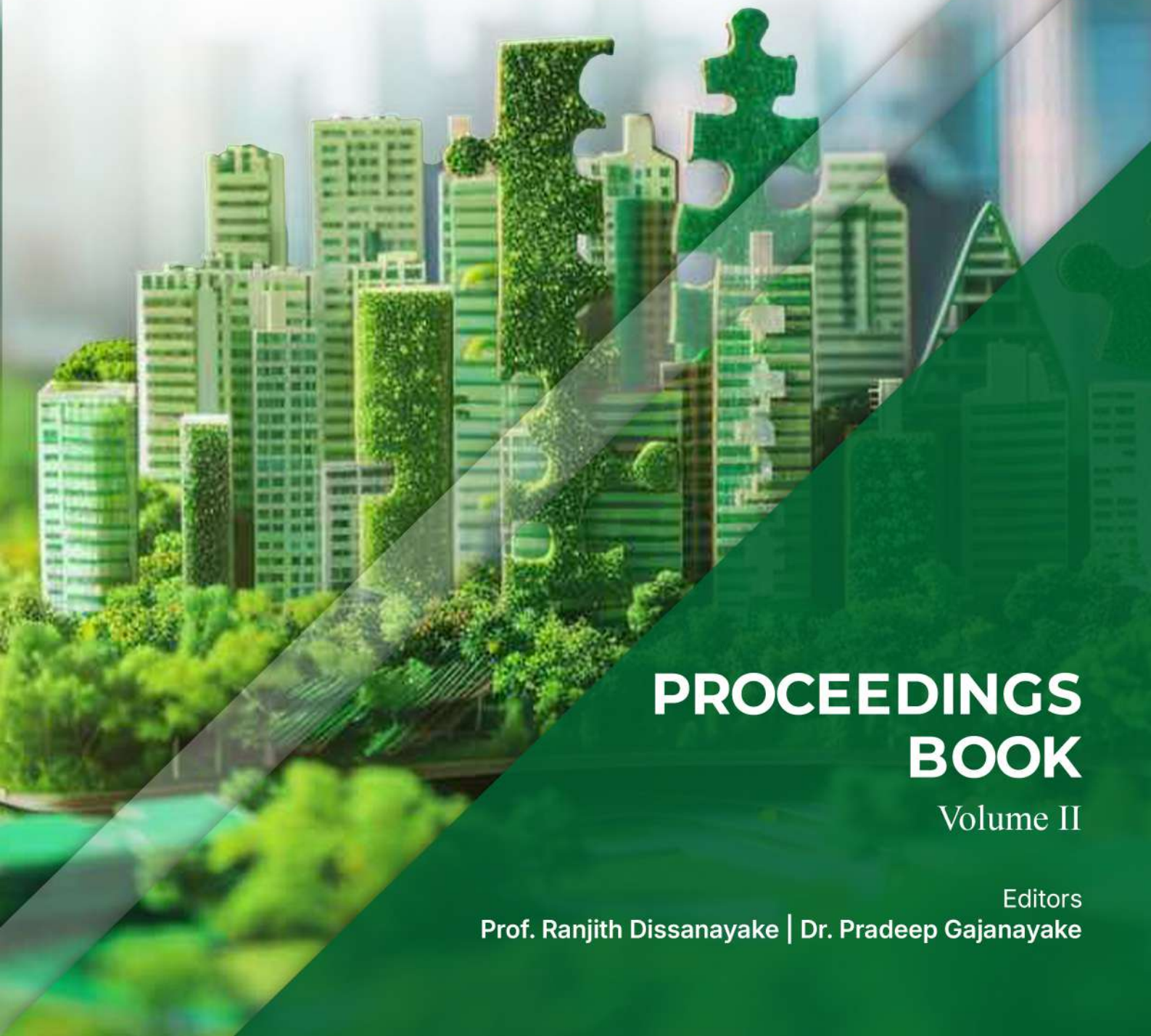




The 16th International Conference on
**Sustainable Built Environment 2025 and
Next-Gen Innovation & Advancement
(DIAMOND 75)**



**PROCEEDINGS
BOOK**

Volume II

Editors
Prof. Ranjith Dissanayake | Dr. Pradeep Gajanayake



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THE ROLE OF STOCK EXCHANGES IN PROMOTING THE BANKING SECTOR IN ADOPTING GREEN FINANCE: A GLOBAL ANALYSIS WITH INSIGHTS FROM EMERGING MARKETS

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Abstract: Over the last 50 years, the field of finance has evolved from three traditional pillars to include a fourth critical pillar to ensure planetary survival, sustainable finance. The urgency to transition from traditional finance to sustainable finance has never been greater, as climate change accelerates, regulatory pressures intensifies globally with developing economies further falling into poverty with increasing biodiversity loss, and investors increasingly face lack of market transparency and demand environmentally and socially responsible portfolios to mitigate long-term risks and secure equitable economic growth. This paper examines the increasing need of cohesion required by stock exchanges, governments and banks in pushing their economies towards following the pathways towards limiting the rise of global temperatures to below 1.5°C and achieving the 2050 goal of Net-Zero. By synthesizing over 64 sources, the study further focuses on scaling sustainable financial instruments and provides an overview of both the financial and sustainability industry. This research study sets an agenda in scaling sustainable finance in the 21st century.

Keywords: Stock exchanges; Sustainable banking; ESG adoption; Climate finance; Emerging markets; Financial regulation

1.0 Introduction

1.1 The roles of stock exchanges

As highlighted by Meng, Boule, & Giuliani (2017), exchanges can serve as platforms for disseminating information about green finance instruments, their benefits, and the risks associated with them. This educational role empowers investors to make informed decisions and allocate capital towards sustainable projects. As shown in Figure 1, projected global GHG emissions based on NDCs announced prior to UNFCCC COP26 indicate that current commitments would still make it likely that global warming exceeds 1.5°C, underscoring the urgency of the awareness-raising role exchanges can play (IPCC, 2023). Figure 2 further illustrates that every region faces increasingly severe and compound climate risks in the near term, reinforcing the need for stock exchanges to disseminate timely and accurate information to investors and market participants (IPCC, 2023).

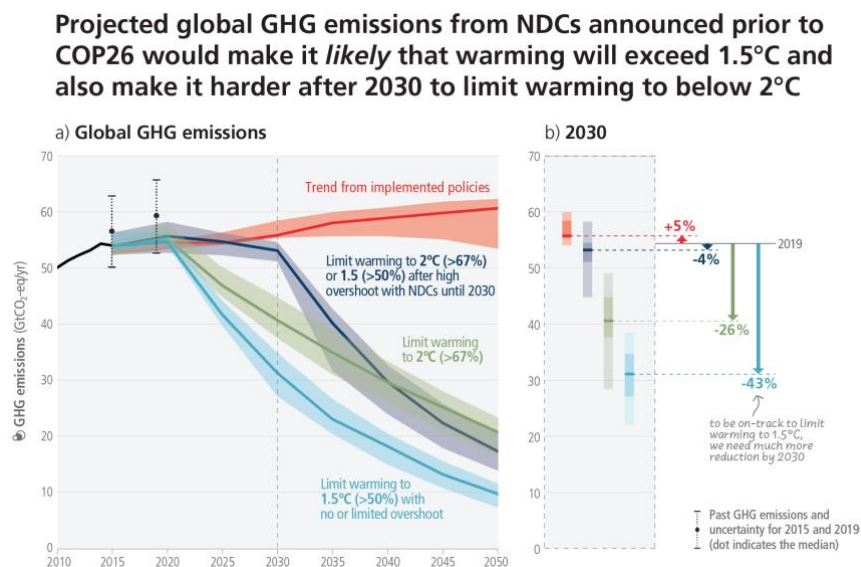


Figure 1 Projected global GHG emissions from NDCs announced prior to UNFCCC COP26 would make it likely that global warming will exceed 1.5°C and make it harder to limit warming to below 2°C. (IPCC, 2023)

Every region faces more severe and/or frequent compound and cascading climate risks

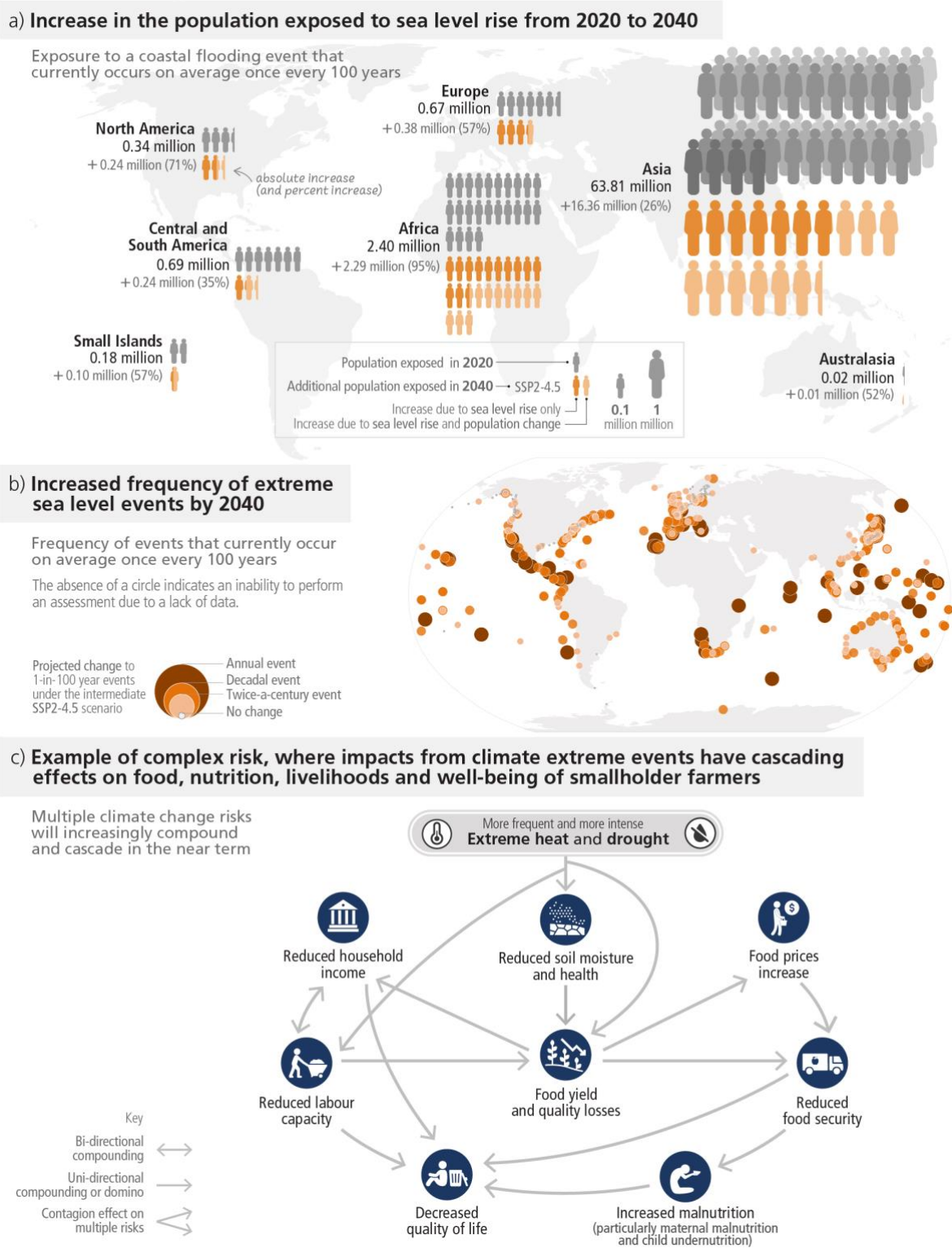


Figure 2 Every region faces more severe frequent compound and/or cascading climate after risks in the near term. (IPCC, 2023)

1.2 Adoption of green finance practices by banks

Banks' direct environmental impact is small compared with financed emissions and stems mainly from power consumption (e.g., data centres). Policies and rules to reduce emissions could raise credit, legal, and reputational risks for banks with large exposures to high-emitting sectors, such as oil and gas, metals and mining, real estate, or transportation. Financing the climate transition positively offers a growth avenue for banks through lending, debt structuring, and other capital markets activities, therefore the sector itself is increasingly integrating green finance practices into its operations although challenges persist. Despite growing literature on green finance, a significant conceptual gap remains in understanding the specific role of stock exchanges in driving banking sector adoption across divergent income contexts, particularly in emerging markets. This study addresses this gap by investigating four interdependent drivers of green finance adoption, market regulations, market mechanisms, awareness, and incentives, through a mixed-methods analysis combining a synthesis of over 64 academic and policy sources with primary data collected from 103 financial practitioners in emerging economies.

Accordingly, this study is guided by the following research question: How can stock exchanges promote green finance adoption by banks across divergent income contexts, and what regulatory, mechanistic, awareness-based, and incentive-driven frameworks are most effective in accelerating this transition in emerging markets? It is hypothesized that the four identified drivers, namely market regulations, market mechanisms, awareness, and incentives, are positively and significantly associated with green finance adoption by banks, with regulatory convergence representing the strongest systemic lever in low- and middle-income economies.

2.0 Methods

2.1 Research design and approach

The focus of the research was more data oriented in order to justify the relationship between how stock exchanges could influence the banking sector and economies and thereby used statistical tools to eliminate research bias (Scotland, 2012; Saunders et al., 2019). A deductive approach was employed, working from a theory to hypotheses to data to add to or contradict

the theory (Creswell, 1998). A case study approach was used for examining the complex, real-world dynamics between multiple emerging economies' stock exchanges and the adoption of green finance within the emerging economies' banking sectors (Yin, 2018; Saunders et al., 2019). This method allowed for in-depth investigation of the stock exchanges within emerging markets' specific context, mechanisms, and influence on banks' green finance practices. While not statistically generalizable, the case study approach offered valuable insights and contributed to broader theoretical understanding of stock exchanges' role in promoting sustainable finance. A cross-sectional time horizon was adopted, providing the collection of data on more than one case and at a single point in time in order to collect a body of quantitative or quantifiable data in connection with two or more variables, which were then examined to detect patterns of relationships (Bryman, 2012).

2.2 Questionnaire design, participants and data analysis

The questionnaire was designed to evaluate how the stock exchanges worldwide influence green finance adoption within the global banking sector, while utilizing stock exchanges within emerging markets as a case study. It was structured around four independent variables including market regulations, market mechanisms, awareness, and incentives, with green finance adoption as the dependent variable. Each factor was broken down into seven unique questions, while the dependent variable had two questions. A 5-point Likert scale was used to measure respondents' answers to the questions provided, from "Strongly Disagree" to "Strongly Agree." This scale was chosen for its simplicity, reliability, and ability to quantify subjective answers provided by various professionals within the financial sector. To ensure clarity and validity, the questionnaire underwent a pilot test with a small group of banking professionals, to identify any changes in wording and ensure authenticity and alignment with research objectives.

The sample population consisted of 144 financial experts within the financial industry with a priority for C-suite position holding individuals within banks and finance-related entities, covering Regulatory Policy and Compliance, Enterprise Risk Management and Post-Trade Settlement, Trading Operations and Market Surveillance in emerging economies. From the sample population size of 144, the corresponding sample was 103 using Krejcie and Morgan's Table. The sampling technique was based on a stratified random sampling method in which

the difference among various financial institutions was used to represent the strata and further, proportional allocation was used by calculating the percentage of respondents from each division. The survey was distributed electronically via Microsoft Forms to 103 banking professionals within the financial industries of multiple emerging economies, representing diverse roles including senior management, C-suite, directors, banking executives and compliance officers across different institutions with a focus on the banking sector. Participants received a consent form explaining the study's purpose, confidentiality measures, and voluntary participation. The survey remained open for 3 weeks, with reminders sent to improve response rates from the banking sector participants.

The results of the research were then taken through SPSS in describing descriptive statistics, including mean and standard deviation, which were used to summarize central tendencies and variability in the data. To assess the internal consistency of the survey instrument, reliability analysis was conducted using Cronbach's alpha. Additionally, correlations (Pearson's r) were used to examine linear relationships between the independent variables and the dependent variable, thus helping to determine which factors had the strongest influence on banks' adoption of sustainable practices. For non-linear relationships, nonparametric correlations (Spearman's test) were applied, which were particularly useful for Likert-scale data, which may not always meet the assumptions of parametric tests. Advanced techniques such as regression analysis were also used to model the combined impact of multiple predictors on green finance adoption. Finally, crosstabs with chi-square tests were used to explore the hypothesis and associations between independent variables.

3.0 Results

3.1 The Impact of Stock Exchange Initiatives on Banks Adoption of Green Finance Practices

The study conducted by Grabski & Miller (2017) highlights how exchanges can facilitate green bond issuance, which in turn encourages banks to participate in green finance initiatives. Figure 3 further indicates the current momentum the global economy has towards pushing for adaptation strategies such as shifting from fossil fuels to renewable energy across onshore wind, offshore wind and photovoltaics through the falling trend of total market costs in

adopting such alternatives. By promoting transparency and facilitating access to green capital, exchanges play a critical role in driving the transition towards a more sustainable financial system. Internal consistency of the primary survey instrument was also confirmed through Cronbach's alpha of 0.841, exceeding the accepted threshold of 0.7.

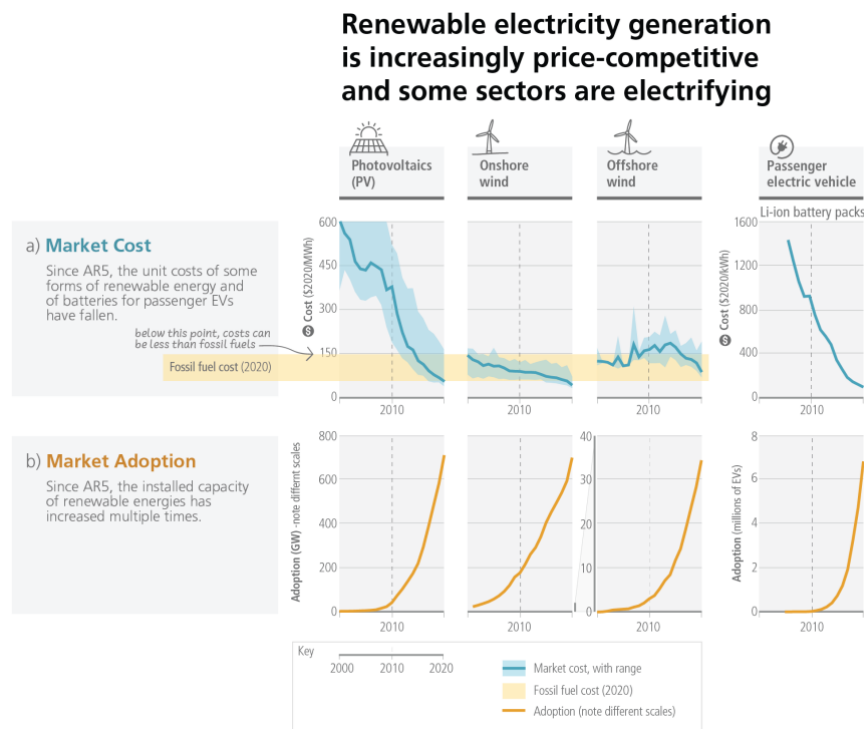


Figure 3 Unit cost reductions and use in rapidly changing mitigation technologies (IPCC, 2023)

3.2 Influence of market regulations

Larsen (2022) points out that the global convergence in green financial policies, spearheaded by influential actors like the EU and China, is a crucial step towards this goal. This convergence leads to harmonized standards and regulations across different jurisdictions, reducing ambiguity and uncertainty for financial institutions and investors. When regulatory frameworks are aligned, it becomes easier for banks and investors to navigate the green finance landscape, assess risks, and make informed investment choices. Furthermore, central banks, as highlighted by Volz (2018), wield significant influence over the financial system. By strategically deploying monetary policy tools, central banks can incentivize environmentally sustainable investments and discourage environmentally harmful activities. These tools can include

targeted lending programs, green bond purchases, and the integration of environmental factors into financial stability assessments. Such regulatory interventions create a favorable environment for green finance by directing capital flows towards sustainable projects and de-risking green investments. In reference to Figure 4, enforcing higher market regulations in highly emitting economies such as North America, Europe and Eastern Asia is dire. As reported from 1850-2019, North America holds a significant 23% contribution to global CO₂ emissions, particularly due to the industrial revolution through the late 1900s. The global convergence in green financial policies, the influence of central banks, and the need for clear regulations governing green financial instruments are all critical factors in driving the transition towards a sustainable financial system.

Emissions have grown in most regions but are distributed unevenly, both in the present day and cumulatively since 1850

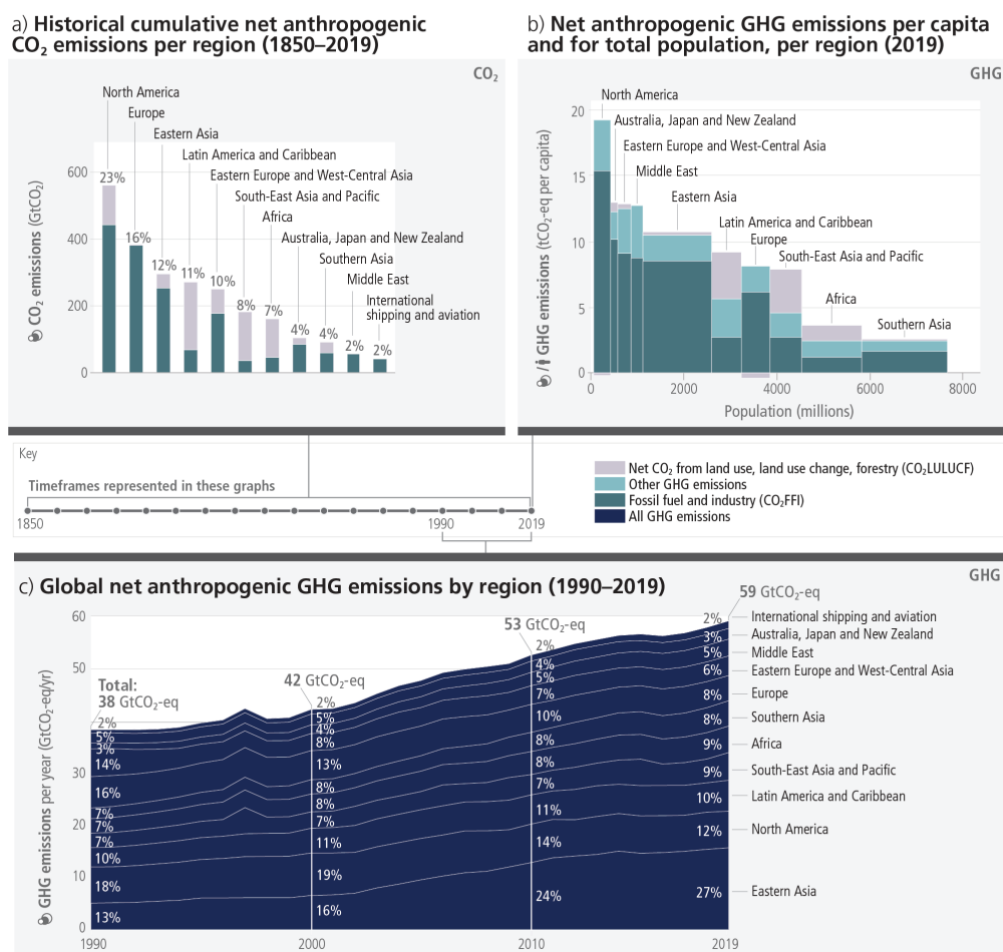


Figure 4 Regional GHG emissions, proportioned CO₂ emissions from 1850-2019 (IPCC, 2023)

Policies set out by governments also play a critical role in shifting approaches market regulators take in channeling finance through the industry. With the US withdrawal from the UNFCCC Paris Agreement in early 2025 and the World Health Organization (WHO), a cascading motion was set in motion, marking what is forecasted to be one of the key turning points of North America regressing from the climate and global commitments (European Parliament, 2025). Indeed, JP Morgan was one of the latest to withdraw from the UN-sponsored Net-Zero Banking Alliance (NZBA), following Bank of America, Morgan Stanley, Wells Fargo, Goldman Sachs and Citigroup, one of NZBA's founding members (Gayle, 2025). Moreover, the growing importance of green bonds and loans in capital markets, as emphasized by Maina et al. (2024), underscores the need for a robust regulatory framework. Clear and consistent regulations governing green financial instruments are essential to ensure transparency, accountability, and investor confidence. These regulations should encompass criteria for green bond issuance, disclosure requirements for green financial products, and mechanisms for verifying the environmental integrity of projects. Primary data collected from 103 financial practitioners also recorded a cumulative mean of 3.52 for market regulations, with chi-square analysis confirming a statistically significant association with green finance adoption ($p < 0.001$).

3.3 Influence of market mechanisms

Anbumozhi & Kimura (2018) shed light on how Industry 4.0, with its advancements in areas like artificial intelligence, the Internet of Things, and robotics, can revolutionize the circular economy. By optimizing resource utilization, reducing waste, and promoting sustainable production models, these technologies can significantly enhance the environmental performance of businesses. This, in turn, makes them more attractive to investors looking for sustainable investments. Furthermore, Zhang, Song, & Nedopil (2024) highlight the critical role of China in driving green finance through innovative market mechanisms. China's policies, such as carbon trading schemes and green bond issuance programs, have not only mobilized significant capital towards green projects but also spurred innovation in the green financial sector. Finally, Zhang & Umair (2023) emphasize the interconnectedness of green finance markets. Their research reveals that these markets, such as green bonds, renewable energy, and carbon trading, exhibit dynamic spillover effects. For instance, the growth of the green bond market can stimulate investment in renewable energy projects, which in turn can increase

demand for carbon credits. The European Green Deal, approved in 2020, is also one of the key global mechanisms adopted in changing the global landscape of sustainable finance. It aims to achieve climate neutrality in Europe by 2050 and to cut greenhouse gas (GHG) emissions by at least 55% by 2030 compared to 1990 levels. One third of the €1.8 trillion investments from the NextGeneration EU Recovery Plan and the EU's seven-year budget will finance the European Green Deal (European Commission, 2024). Figure 5 shows an outline of the EU green deal.

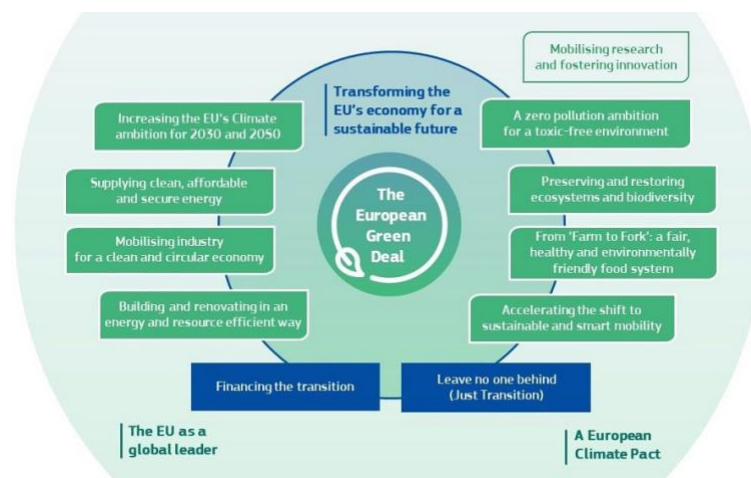


Figure 5 The European Green Deal (European Commission, 2019)

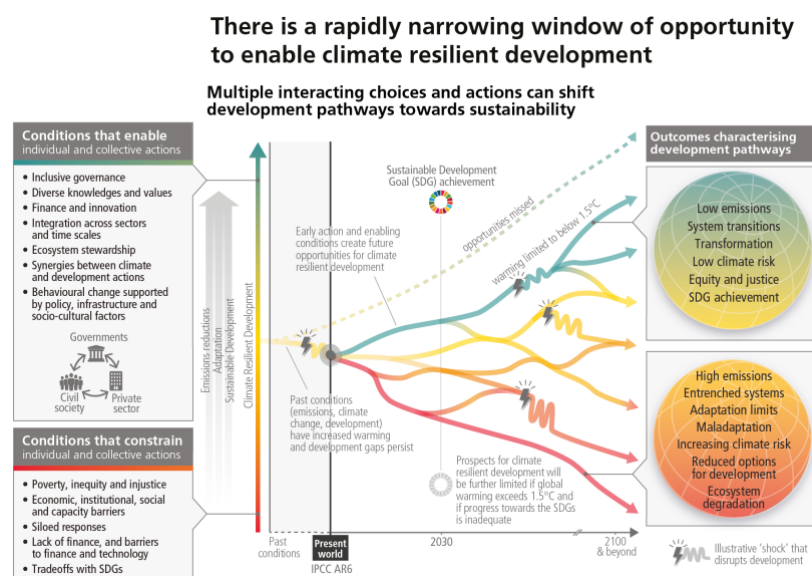


Figure 6 Rapidly closing window of opportunity to enable climate-resilient development (IPCC, 2023)

As part of the European Green Deal and action plan on financing sustainable growth, the European Green Bond Regulation, also referred to as the European Green Bond Standard (EuGBS), establishes a voluntary designation for green bonds which fulfill specific requirements related to the use of proceeds, reporting and disclosure. The designation aims to help direct and scale investment towards sustainable economic activities aligned with the EU's climate and broader environmental goals. For issuers and investors, the designation aims to strengthen the integrity, transparency and level of comparability of the sustainable bond market by providing clear definitions of what green means, in line with the EU Taxonomy, and standardizing reporting and disclosure requirements. Figure 6 further showcases that there is a rapidly narrowing window of opportunity to replicate such mechanisms employed by the European Union (EU) in other regions to enable climate-resilient development. Overall, primary findings also recorded a cumulative mean of 4.04 for market mechanisms, with chi-square analysis confirming a statistically significant association with green finance adoption ($p < 0.001$).

3.4 Influence of awareness

Awareness plays a pivotal role in fostering the growth of green finance by building trust, encouraging participation, and empowering stakeholders to make informed decisions. Transparency and disclosure are key elements in this process. Tröger and Steuer (2021) highlight the significance of disclosure in Europe, emphasizing how it provides investors with crucial information about companies' environmental performance. By increasing transparency, disclosure helps build trust between investors and companies, fostering a more sustainable and responsible financial system. Moreover, awareness campaigns play a crucial role in driving both investor and institutional interest in green finance. Schaefer (2017) focuses on the German banking sector, demonstrating how awareness campaigns have successfully raised awareness about the benefits of green finance among both investors and financial institutions. By showcasing the financial and environmental benefits of green investments, awareness campaigns encourage both individual and institutional investors to participate in the green finance market. Furthermore, the role of stock exchanges in promoting sustainability cannot be underestimated. Myklebust (2013) highlights how the Oslo Stock Exchange in Norway has actively promoted sustainability among listed companies through awareness campaigns and

other initiatives. By setting clear sustainability standards and encouraging companies to disclose their environmental and social performance, the Oslo Stock Exchange has played a crucial role in shaping market practices. SGX, a key exchange within the region, has heavily separated itself as a market leader in the stock exchange industry in Asia by being one of the first and largest exchanges to provide five sustainable-finance-linked product categories across securities, fixed income, equity derivatives, commodities and indices. As shown in Figure 7, SGX also provides five separate fixed-income sustainability-linked securities on its exchange, two of which stand out as social bonds and transition bonds.



Figure 7 Sustainable Finance Fixed Income Securities listed on SGX (Singapore Exchange, 2025b)

Social bonds provide financing for social projects across affordable basic infrastructure, access to essential services, which in turn aids in reducing the number of people living below the poverty line, and protection of marginalized communities, people with disabilities, youth and other vulnerable groups such as indigenous people as noted by the International Capital Market Association (ICMA, 2020). The implementation of transitional bonds, especially in both developed and developing economies, aids in converting high-emitting industries to low-carbon-intensive industries. Figure 8 outlining SGX Group's climate transition plan for 2026

onwards and Figure 9 outlining practices undertaken by CSE also further show the need for urgent collaboration, transparency and relative market education of sustainability by and for stock exchanges. In contrast to the actions undertaken by SGX under its sustainability transition plan, CSE fails to encompass a competitive sustainability landscape. With ‘entry-level’ information disclosed under their sustainability reporting standards in comparison to SGX’s action plan towards achieving Science-based targets under the Science Based Targets Initiative (SBTi), it is evident that exchanges in developing nations affected by the convergence theory according to Kerr (1983), such as CSE, are in dire need of being pushed by entities such as the World Federation of Exchanges (WFE) in regional collaborations specifically in terms of technology and knowledge transfer. This is further supported by CSE’s reporting, which notes that eight memoranda of understanding (MOUs) have been signed with eight regional exchanges under SDG 17 - Partnerships for Development.

Foundation	Implementation Strategy	Engagement Strategy	Governance
Objectives and priorities - To reach net-zero by 2050 - Support the net-zero transition of our operations, marketplace and ecosystem through three key strategic priorities	Strategic priority 1: SGX Group's own decarbonisation and emissions reduction Implementing our science-based targets and reporting progress in our sustainability report since FY2021 Strategic priority 2: Facilitate credible transition of our marketplace and support capital mobilisation through sustainable/ transition products and solutions as a trusted infrastructure provider	Strategic priority 3: Market stewardship and influence through ecosystem collaboration Internal - Regular and accessible climate and sustainability-related education and training for staff to implement SGX Group's climate priorities and policies External - SGX Group's capacity development initiatives involve active engagement with a broad range of ecosystem stakeholders to promote climate-related disclosures and advance climate actions	Robust governance structures across various levels are in place and foundational at SGX Group for driving our climate and sustainability ambitions and actions
Metrics and Targets			
Focus areas: Exchange-focused targets - Emissions reduction and net-zero targets - Maintain robust governance structure and adequate staff training to support our climate actions Market-focused targets Transparency and education: Provide training, awareness activities and guidance to encourage market participants to advance their climate-related disclosures and promote net-zero commitments Products and services: Promote adoption of climate focused products, solutions and platforms. Collaborate with ecosystem partners to develop products and solutions that incorporate climate and sustainability considerations Policy, persuasion and engagement: Proactive engagement with stakeholders, including market participants, partners, policy makers and standard setters, to advance net-zero commitments			

Figure 8 SGX Group’s Climate Transition Plan (Singapore Exchange, 2025a)

Presented below are a few of the initiatives implemented under our ESG strategy, which is aligned with the objectives of the United Nations Sustainable Development Goals (SDGs)

	• Donation of Rs.25 Mn to the Rebuild Sri Lanka Fund • CSE supported national flood relief efforts by donating dry rations as part of our commitment to social responsibility • Conducted a wellness campaign for employees, alongside a blood donation drive
	• Conducted capital market awareness programmes under the theme 'Path to a Public Listing' at University of Sri Jayewardenepura and South Eastern University of Sri Lanka. • Sponsored J/PURA EXPO 2025 and conducted a segment on financial literacy and capital market opportunities • Offered the first-ever, university-industry collaborative Diploma, offered by Rajarata University of Sri Lanka in partnership with the CSE • Conducted capacity-building programmes on ESG for listed entities • Developed several new courses, expanding the academic offerings of the CSE Academy • Opened a new branch in Bandarawela • 4 courses were offered by CSE Education Academy for the year 2025 and 1,155 students attended these courses.
	• Ring the bell for gender equality. • Ensured a female representation of 48%, underscoring efforts to create a supportive and safe workplace for women.
	• ASPI growth of 41.89% • One of the leading Stock Exchanges in the region • Rs. 123 Bn of Capital raised through the CSE by listed companies. • Foreign capital inflows of Rs. 76 Mn • Empowering SMEs on Capital Raising through Listing • Successfully concluded the 'Invest Sri Lanka' Investor Forum in Singapore

Figure 9 An extract from the 2025 annual report submitted by the CSE
(Colombo Stock Exchange, 2025)

In conclusion, awareness plays a multifaceted role in driving the growth of green finance. By increasing transparency, empowering stakeholders, and building trust, awareness initiatives create a conducive environment for the development of a sustainable and responsible financial system. As Tröger and Steuer (2021), Schaefer (2017), and Myklebust (2013) demonstrate, disclosure, awareness campaigns, and the active engagement of stock exchanges are crucial for fostering widespread adoption of green finance practices. Primary findings also recorded a cumulative mean of 4.04 for market awareness, with chi-square analysis confirming a statistically significant association with green finance adoption ($p < 0.05$).

3.5 Influence of incentives

Incentives can take various forms, including financial subsidies, tax breaks, and regulatory benefits, all aimed at offsetting the initial costs associated with green investments and

encouraging innovation. The research conducted by Naeem et al. (2022) provides valuable insights into the interconnectedness between green finance assets and sectoral stock markets. Their findings reveal strong correlations, indicating that investors are increasingly drawn to sectors that incorporate green finance principles. Furthermore, Patel & Jose (2024) shed light on the significant impact of tax incentives on corporate accounting practices in the United States. Their research demonstrates that these incentives can make green investments more appealing to businesses by reducing their tax liabilities. By incentivizing sustainable practices, these tax benefits can encourage banks to adopt greener technologies and invest in environmentally friendly projects. The need for a global approach to incentivizing green finance is emphasized by Niyazbekova et al. (2021) whose research highlighted the importance of international collaboration in designing and implementing effective incentive structures. By sharing best practices and coordinating efforts, stock exchanges worldwide can create a more favorable environment for green finance adoption on a global scale. Non-financial incentive frameworks also play a key role alongside financial incentive frameworks provided by stock exchanges, in being key drivers in pushing banks and other entities to execute green and social financial instruments within their markets. The case studies represented in Tables 1, 2 and 3 showcase the improvements made by stock exchanges globally towards incentivizing green investments in accordance with the guidelines set out by the World Federation of Exchanges Green Equity Principles. These incentivizing mechanisms provide entities executing large-scale action plans with recognition within their respective stock exchange.

Table 1 Nasdaq Green Designation. Source: Author

In June 2021, Nasdaq launched voluntary Green Designations targeting green and transition companies on Nasdaq Nordic markets. The Nasdaq Green Designations provide transparency on the green credentials of a company and offer a way to follow a company's progress over time. To meet the Green Equity Designation principles, companies must have more than 50 percent of turnover from green activities and continue to invest in a majority share of green activities, in addition to providing transparency on EU Taxonomy alignment and company-level sustainability targets (S&P Global Ratings, 2025a).
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Table 2 SIX 1.5 °C Climate Equity Flag Designation. Source: Author

In August 2024, the SIX Swiss Exchange launched the SIX 1.5 °C Climate Equity Flag, which helps companies provide additional supporting evidence that their entire value chain contributes towards limiting global warming to 1.5 °C above pre-industrial level. To meet the SIX 1.5°C Climate Equity Flag requirement, more than 50 percent of the issuer’s annual revenues must come from 1.5°C-aligned activities (S&P Global Ratings, 2025a).

Table 3 B3 Green Equities (BAV). Source: Author

In May 2024, B3, the Brazilian Stock Exchange launched a voluntary B3 Green Equities (BAV) designation targeting green companies in Brazilian markets, based on the World Federation of Exchanges Green Equity Principles. To meet the Green Equity Designation principles, companies must have more than 50 percent of annual gross revenue from activities that contribute to the green economy and continue to invest in a majority share of green activities (S&P Global Ratings, 2025a).

Further analysis is also shown by S&P through their award-winning Shades of Green approach, attached in Figure 10, which shows the expected alignment of a company’s activities with a low-carbon, climate resilient future (and alignment with the Paris Agreement), based on the feasible transition timeline for the company’s sector and its own transition plan/commitments.

S&P Global Ratings’ Shades of Green

Assessments					
 Dark green	 Medium green	 Light green	 Yellow	 Orange	 Red
Description					
Activities that correspond to the long-term vision of an LCCR future.	Activities that represent significant steps towards an LCCR future but will require further improvement to be long-term LCCR solutions.	Activities representing transition steps in the near-term that avoid emissions lock-in but do not represent long-term LCCR solutions.	Activities that do not have a material impact on the transition to an LCCR future, or, Activities that have some potential inconsistency with the transition to an LCCR future, albeit tempered by existing transition measures.	Activities that are not currently consistent with the transition to an LCCR future. These include activities with moderate potential for emissions lock-in and risk of stranded assets.	Activities that are inconsistent with, and likely to impede, the transition required to achieve the long-term LCCR future. These activities have the highest emissions intensity, with the most potential for emissions lock-in and risk of stranded assets.
Example projects					
 Wind power	 Certified forestry	 New energy efficient buildings	 Fossil fuel buses	 Conventional steel production	 Oil and gas exploration

Figure 10 S&P Global Ratings’ Shades of Green approach (S&P Global Ratings, 2025a)

The Second Party Opinion (SPO), provided by S&P Global Ratings, has also proven to be one of the key rating agency tools in the financial industry, driving green finance markets worldwide with a proven case study from the Nordic Investment Bank (NIB) as in Table 4 and Banco de Desarrollo del Ecuador B.P. in Table 5. In conclusion, these studies collectively underscore the critical role of incentives in promoting green finance in both developing economies and developed economies, particularly in South Asia, Africa and South America. Thus, stock exchanges, market regulators, policy makers and national governments should play the leading role in creating a pathway towards incentivizing financial entities such as banks and working together to create a more sustainable financial system. Primary findings also recorded a cumulative mean of 3.99 for incentives, with chi-square analysis confirming a statistically significant association with green finance adoption ($p < 0.001$).

Table 4 Case study of how ratings provided a favorable outcome for Nordic Investment Bank. Source: Author

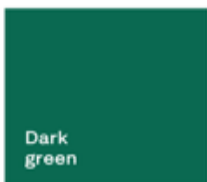
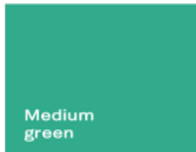
 Activities that correspond to the long-term vision of a low-carbon climate resilient future. Our Shades of Green Analytical Approach >	Nordic Investment Bank (NIB) <u>The challenge</u> The Nordic Investment Bank (NIB) is an international financial institution owned by eight member countries. In 2011, NIB developed its Environmental Bond Framework (NEB framework), which sets out clear criteria for project selection, governance, management of proceeds, and transparent impact reporting. Under this framework, the bank issued Environmental Bonds and Blue Bonds whose proceeds are used to finance projects that benefit the environment and support the transition to a low-carbon economy. <u>The solution - Non-financial recognition through ratings</u> To provide investors with greater transparency into its environmental bond framework and to help demonstrate its commitment to maintaining high environmental standards in its green financing initiatives, NIB secured a Second Party Opinion (SPO) from S&P Global Ratings. S&P Global Ratings confirmed that NIB's environmental bond framework is fully aligned with the Green Bond Principles (GBP). S&P Global Ratings also assessed NIB's framework as Dark green according to its Shades of Green methodology, confirming that NIB's activities correspond to the long-term vision of a low-carbon, climate-resilient future. <u>The outcome</u> Following a green bond framework update in 2024, NIB issued a five-year €750 million NIB environmental bond, the largest green bond in the bank's history. The transaction attracted strong investor demand with a broad distribution across geographies and investor types. Overall, 2024 stands as a record year for NEB issuance. NIB has aligned its framework with recognized market standards like the GBP and the EU Taxonomy, and has secured independent assessments such as the SPO, to help achieve its objective of attracting environmentally conscious investors (S&P Global Ratings, 2024).
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Table 5 Case study of how ratings provided a first step towards scaling green and social finance for Banco de Desarrollo del Ecuador B.P. (BDE B.P.). Source: Author

 Medium green Activities that represent significant steps towards a low-carbon climate resilient future but will require further improvements to be long-term low-carbon climate resilient solutions. Our Shades of Green Analytical Approach >	<u>Introduction</u> Banco de Desarrollo del Ecuador B.P. (BDE B.P.) is a government-owned financial institution established in 1979. It primarily offers financing and technical assistance to Ecuador's 93 decentralized autonomous governments for infrastructure and public service projects. The bank's activities encompass sectors such as transportation, water and sanitation, urban development, and housing. It is headquartered in Quito, Ecuador. <u>The solution</u> S&P Global provided a Second Party Opinion for Banco de Desarrollo del Ecuador B.P.'s Green Bond Framework on May 7, 2025, providing recognition for the bank to drive Foreign Direct Investment (FDI) into its economy and take a first step towards driving investments into green and social markets within the country. The SPO granted the bank “Medium Green” under its Shades of Green approach, highlighting that the bank’s activities represent significant steps towards a low-carbon, climate-resilient future but will require further improvements to be long-term, low-carbon, climate-resilient solutions. Thus, the case study further suggests that the first initiative should be taken by financial regulators in developing economies worldwide, in this case a government-owned financial institution in South America, in creating a livable environment for its people as well as the environment and tackling climate change (S&P Global Ratings, 2025b).
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4.0 Discussion

The urgent need for improving regulations across markets is dire to support green finance adoption. According to a study conducted by the International Monetary Fund and the International Organization of Securities Commissions, regions show significant differences in the level of implementation of sufficient market regulations, which, to a certain extent, are correlated with the overall income level of the region. Europe and Asia exhibit the highest levels of implementation, the Western Hemisphere ranks in the middle, while the Middle East and Central Asia and Africa exhibit the lowest levels of implementation as shown in Table 6 and Figure 11. Table 7 and Figure 12 further signify this study as only 49% of the average share of regulators from low-income countries has supportive market regulations in their economy while high-income countries that are part of the Organisation for Economic Co-operation and Development (OECD) accumulate to a staggering 85% supportive regulations implemented within their nations.

Table 6 IOSCO Categories and Countries Grouped by Region (IMF, 2007)

Region	Regulator	SRO	Enforcement	Cooperation	Issuers	CIS	Market Intermed.	Secondary Market
Africa	50	56	62	46	67	72	39	46
Asia and Pacific	82	68	67	63	77	68	64	72
Europe	74	51	72	77	68	86	69	82
Mid. East Central Asia	59	44	36	21	47	40	56	48
Western Hemisphere	64	50	52	39	61	64	60	53

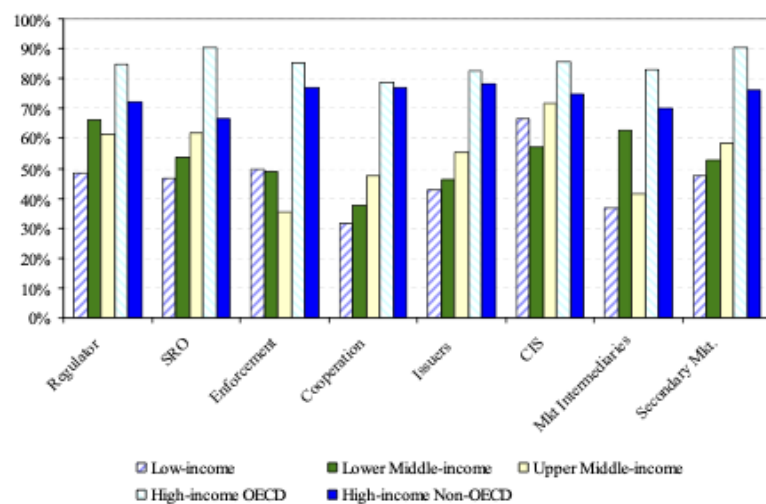


Figure 11 IOSCO Categories and Countries Grouped by Region (IMF, 2007)

Table 7 IOSCO Categories and Countries Grouped by Income (IMF, 2007)

Category	Low-income	Lower Middle-income	Upper Middle-income	High-income OECD	High-income Non-OECD
Regulator	49	66	61	85	73
SRO	47	54	62	91	67
Enforcement	50	49	36	85	77
Cooperation	32	38	48	79	77
Issuers	43	47	56	83	79
CIS	67	57	72	86	75
Mkt Intermediaries	37	63	42	83	70
Secondary Mkt.	48	53	59	91	76

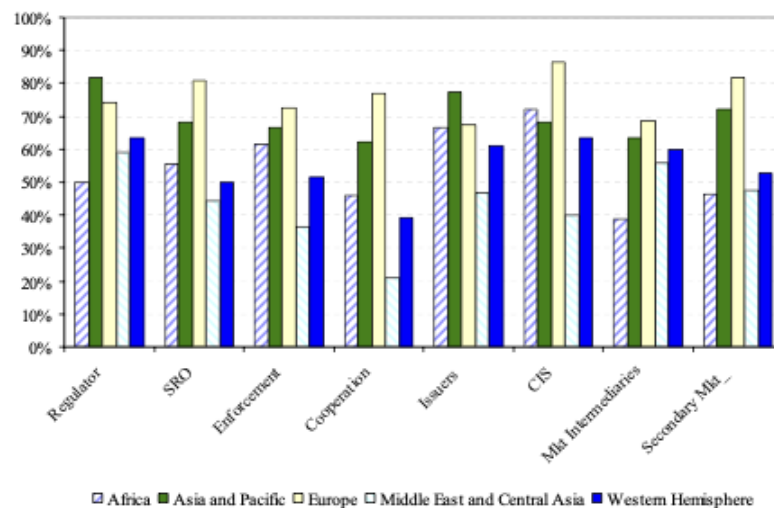


Figure 12 IOSCO Categories and Countries Grouped by Income (IMF, 2007)

Entities, such as banks, are solely influenced by profit incentives, which are thereby supported if there is a large-scale demand shift for green and social financial products by their consumers. To ensure actionable outcomes, the following SMART framework under Table 8 could be utilized to solidify the relationship between market regulations and green finance adoption by banks. Figure 13 further illustrates the range of opportunities available for scaling up climate action across markets (IPCC, 2023).

There are multiple opportunities for scaling up climate action

a) Feasibility of climate responses and adaptation, and potential of mitigation options in the near-term

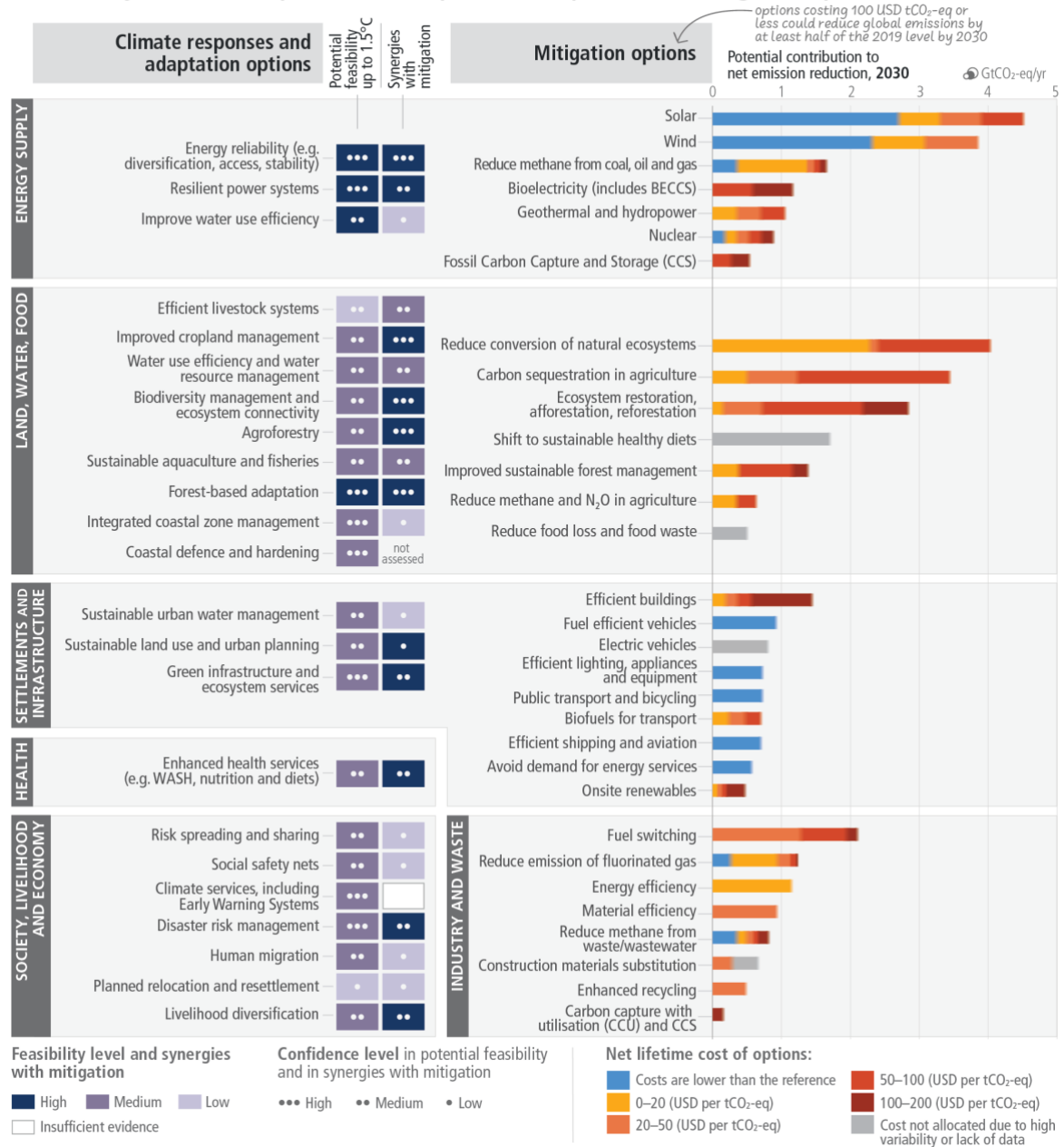


Figure 13 Multiple Opportunities for market scaling up climate action. (IPCC, 2023)

Table 8 SMART for improving the relationship between regulations and green finance adoption. Source: Author

SMART Criteria	Implementation Plan
Specific	Mandate climate risk disclosure rules for all licensed banks with their economy requiring;
	1. Annual detailed reporting of financed emissions (Scope 3).
	2. Public disclosure of green asset ratios (GAR) by Q2 of Year 2: 2028.
Measurable	Enforcement benchmarks include;
	1. 100% compliance with disclosure rules by Q2 of Year 2: 2028.
	2. 40% of bank portfolios allocated to green and social financing loans by Q3 of Year 3: 2029.
	3. Penalties for non-compliance with the regulations set (e.g., 1.5% of revenue for missed targets).
Achievable	Proposed regulatory partnerships include;
	1. Central Bank of the emerging economy to adopt European Central Bank (ECB) linked climate stress tests.
	2. Security and Exchange Commission of the emerging economy to align with regional associations/bodies of sustainability promoters of ESG disclosure standards within the banking sector (e.g., ASEAN).
	3. Technical assistance from IMF's Climate Finance Task Force.
Relevant	Linking targets to IPCC's 1.5°C pathways (e.g., 40% emissions cuts in transport/buildings by 2030)
Time-bound	Phase 1 (Q4 of Initial Year: 2026): Draft regulations finalized
	Phase 2 (Q4 of Year 1: 2027): Pilot the regulations with systemically important/larger banks within the economy as per market share/revenue.
	Phase 3 (Q4 of Year 2: 2028): Full enforcement of the regulations

Stock exchanges in emerging economies should adopt strategic frameworks similar to that of the United Nations Sustainable Stock Exchanges (SSE) initiative's model for carbon market expansion (UNSSE, 2023) in order to enhance synergy between market mechanisms and the expansion of green finance. This centralized governance approach is particularly important for developing economies such as Sri Lanka, which often face significant delays in integrating global financial innovations due to its high debt-to-GDP ratio (UNSSE, 2023). The Colombo Stock Exchange (CSE), for example, experienced a 15-year gestation period before issuing its inaugural green bond in late 2024, despite the asset class originating globally in 2007 via the

European Investment Bank (EIB). As illustrated in Figure 14, empirical policy research, multi-stakeholder consensus-building, and the provision of technical advisory services focused on scaling carbon markets through exchange-led frameworks act as three fundamental pillars with regard to the UN SSE (UNSSE, 2023).



Figure 14 SSE action framework for maximizing opportunities of carbon markets (UNSSE, 2023)

Table 9 SMART for improving the relationship between market mechanisms and green finance adoption. Source: Author

SMART Criteria	Implementation Plan
Specific	Establish a dedicated Green Finance Task Force at emerging economies' stock exchange to implement UN SSE's Carbon Market Framework, focusing on raising standards for ESG disclosure requirements for listed companies, while projecting an action plan showcasing a rule for every bank to list a green financial product by 2030.
Measurable	Set quantifiable targets such as meeting 10+ certified green bonds listed by Q2 of 2028 and launching a pilot carbon credit trading platform by Q3 of 2028 together with the national government.
Achievable	Leverage existing partnerships and resources with the Securities and Exchange Commission (SEC) in the host emerging economy and the Ministry of Environment, Multilateral Development Banks (MDBs) and the national government through Memoranda of Understanding (MOUs).
Relevant	Align with host nation's updated Nationally Determined Contributions (NDCs), 2050 Net-Zero Roadmap and the UN Sustainable Development Goals (SDGs) 7, 12, and 13.
Time-bound	Phase 1 (Q4 of Year 1: 2027): Develop a new/updated Regulatory Framework.
	Phase 2 (Q1 of Year 2: 2028): Complete a Market Readiness Assessment within the emerging economy.
	Phase 3 (Q2 of Year 3: 2029): Operationalized Full Implementation to achieve various sustainability goals, including that of 2030 for each listed bank to issue green instruments.

To strengthen the link between market mechanisms and green finance adoption, two primary objectives are proposed in Table 9;

1) Market Integrity and Expansion acts as Objective 1 as it focuses on the systemic drivers that allow exchanges to grow the scale and transparency of green and social markets.

2) Developing a credible green and social market offering acts as Objective 2 as it guides exchanges in establishing credible, localized green and social market offerings. Ensuring market functionality, maintaining stability, and creating a reliable pipeline for sustainable finance products act as three key pillars with regard to achieving this objective (UNSSSE, 2023).

To ensure actionable outcomes, the following SMART framework under Table 10 could be utilized to solidify the relationship between market awareness and green finance adoption by banks.

There exists a need for greater information exchange and knowledge building in terms of green and social finance markets. Exchanges can contribute to capacity-building efforts through their own initiatives as well as through supporting efforts of third parties such as rating agencies globally, as well as their fellow exchanges who are moving forward faster with green finance initiatives across their respective regions. For example, the Colombo Stock Exchange (CSE) could partner with the Singapore Exchange (SGX) in learning through peer-to-peer meetings to overcome any obstacles which would be present in adopting green finance while developing guidance on best practices, rather than primarily focusing on building their own exchange nationally. Similarly, in reference to the example provided in Figure 15 of a collaboration formed by NASDAQ and a Verified Carbon Market (VCM) entity, collaborations between exchanges in emerging economies and developed economies such as the Luxembourg Stock Exchange (LuxSE) may be subject to agreements among exchanges and could be highly regulated, yet they may also benefit from stakeholder engagement activities in the future, while driving green markets forward faster and securely.

Box C.2: Nasdaq's varied involvement in the VCM environment

Nasdaq, Inc. (Nasdaq) is engaged with the VCM ecosystem at multiple points of the value chain. Since 2021, Nasdaq is the majority owner of Puro.earth Oy ("Puro"), a standard setting body for engineered carbon removal methods in VCMs. Puro is an integrated standard setter and registry focusing on high quality carbon removals with long durability (minimum one hundred years) and low risk of reversal. Through the Puro Standard, Carbon Removal Credits ("CORCs") are issued, each representing one metric ton of carbon removed from the atmosphere. CORCs are only issued once the carbon removal process has been completed, they are not based on a projected removal that is expected to take place at some point in the future. CORCs are issued in the Puro Registry where their complete lifecycle is recorded, from issuance to retirement.

In 2022, Nasdaq launched commodity reference price indices based on the trading of Puro's CORCs. These indices aim to bring price transparency to the market and help corporations understand the true cost of removing their carbon emissions. Additionally, Nasdaq Marketplace Services Platform provides carbon credit exchanges with trading technology designed to match buyers and sellers with carbon credits based on their specific requirements. In 2022, Nasdaq and Climate Impact X (CIX) entered into a partnership, leveraging Nasdaq's cloud-based trading technology to power CIX's new spot exchange platform.

Figure 15 Collaboration between (NASDAQ) and Puro.earth Oy (ICVCM, 2024)

Table 10 SMART for improving the relationship between market awareness and green finance adoption. Source: Author

SMART Criteria	Implementation Plan
Specific	Formalize co-regional technology transfer mechanisms within emerging economies and/or developed economies (e.g., Colombo Stock Exchange and Singapore Exchange) through working groups on green bond listings and carbon market integration.
Measurable	Target adoption of three developed-economy-aligned ESG policies within the first 12 months through the emerging market stock exchange.
Achievable	Leverage UN SSE partnerships with global exchanges and entities to raise funding and technical support.
Relevant	Align with domestic country's national climate strategy (2023-2030), in accordance with updated NDC targets.
Time-bound	Phase 1 (Q3 of Initial Year: 2026): Policy drafting
	Phase 2 (Q1 of Year 1: 2027): Launch
	Phase 3 (Q1 of Year 3: 2028): Expansion to the region

One of the key requirements of exchanges is promoting the availability of decision-useful information for investors and other market users such as banks and governments. According to the latest data, so much progress has been made in sustainability-related/ESG disclosures worldwide, yet in most emerging markets you can clearly see an absence of such availability. Raising market awareness also calls for greater transparency as exchanges are well positioned to promote the quality of available data and reduce information asymmetry by promoting better disclosure practices around green and social markets. In Voluntary Carbon Markets (VCMs) particularly, the lack of access to project-level detail and the scarcity of pricing data of each

credit limit the ability of buyers to perform due diligence for effective price discovery. By promoting standardized disclosures on green and social finance market activity, exchanges can afford stakeholders a better view of the use of carbon market instruments. Equally, inadequate disclosure about green finance market activity relative to actual emission reductions undermines the credibility of reduction claims by banks and companies and poses reputational risk to the green finance markets in question (UNSSE, 2023).

With the most climate-affected region, the Asia-Pacific, being the third-largest region in green bond issuance with over USD 219.3bn issued as of 2020 while Europe and North America accounted for over USD 650 billion in issuance according to the Climate Bonds Initiative (CBI), the urgent need for incentivizing financial markets across the region can no longer be ignored. The need for incentivizing the adoption of green finance is further proved in Figure 16 with emerging economies and supranational economies only accounting for 28% of green bond issuance while developed economies account for over 72%.



Figure 16 Country allocation of green bond execution (Climate Bonds Initiative, 2020)

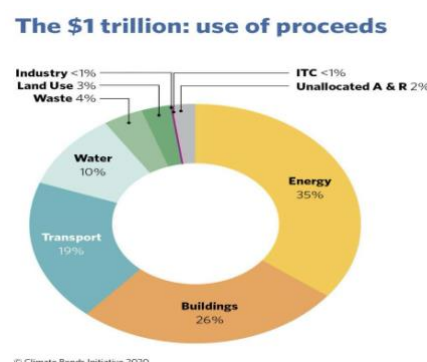


Figure 17 Use of proceeds by green bonds (Climate Bonds Initiative, 2020)

Thus, the recommendation of how the relationship between incentives and green finance adoption may rely on two significant questions; who should market regulators and stock exchanges prioritize in incentivizing first and how industries should be incentivized. Globally, investment in the Energy Sector comprises the largest component of the multi-billion-dollar green bond issuances, at USD 354.7bn, followed by Low-carbon Buildings with USD 263.5bn and Transport in third place with USD 190.7bn issued. Thus, as shown in Figure 17, it could be argued that industries such as energy, buildings and transport should further be incentivized, while industries such as land use should not go unnoticed. Rainforest conservation, primarily governed by the United Nations Convention on Biological Diversity (UN CBD) and prevention of land degradation, primarily governed by the United Nations Convention to Combat Desertification (UNCCD), should be the industries primarily prioritized in incentivizing as to the lack of awareness and education surrounding the importance of such industries. Although mechanisms such as the UN-employed Clean Development Mechanism (CDM) under Article 6 support developments through carbon credits, the latest UNFCCC COP30 was unable to provide a clear global consensus behind carbon pricing, though it did finally reach agreement on carbon trading under Article 6, bringing nearly a decade of negotiations to a close (UNFCCC, 2025). With regard to recommending a solution pertaining to how market regulators and stock exchanges should incentivize green finance adoption by banks and other entities, it could be noted that going back to fundamental methods of financial incentives such as tax breaks and subsidizing and non-financial incentives such as recognition for entities within exchanges for their transition plans and adoption is the best way to move forward, such as the Sri Lanka Electricity Act, No. 36 of 2024 which calls for market-based incentives to promote renewable energy within the nation (POTDSL, 2024). Retracing to initiatives such as the SPOs and Climate Transition Assessments (CTAs) set out by rating agencies such as S&P Global in collaboration with exchanges such as NASDAQ, B3 and SIX Swiss Exchange and scaling them to emerging economies and showing recognition for listed entities adopting such initiatives is also important. This also excludes any solutions that do not support such practices, such as heavy market regulations or high taxation in adopting such mechanisms as means to acquire higher government tax revenue, eliminating the sole purpose of financing environmental conservation and social governance. To ensure actionable outcomes, the following SMART framework under Table 11 could be utilized to solidify the relationship between incentives and green finance adoption by banks.

Table 11 SMART framework for improving the relationship between incentives and degree of green finance adoption. Source: Author

SMART Criteria	Implementation Plan Details
Specific	Targeted industry incentives for Energy, Low-Carbon Buildings, Transport, and Land Use (Rainforest Conservation and Soil Enrichment). Includes a mandate for 20% of bank portfolios to be directed toward these sectors by Year 2: 2028.
Measurable	Achieve 30% of stock-exchange-listed entities adopting IPCC's 1.5°C-aligned transition plans by 2027.
Achievable	Collaborate with the Security and Exchange Commissions and the Departments of Inland Revenue to execute tax breaks for green bonds.
Relevant	Align with global goals, including UNFCCC Article 6 carbon trading and the 2030 Agenda for Sustainable Development, while linking incentives to IPCC 1.5°C pathways.
Time-bound	Phase 1 (Q1 of Year 2: 2028): Pilot tax breaks within the emerging nation.
	Phase 2 (Q1 of Year 3: 2029): Expand green bond issuance to 35% of listed bond issuances within the stock exchange.

This study also carries certain limitations including the fact that the primary data was collected exclusively from financial practitioners in emerging economies including but not limited to India, Malaysia, Ghana and Sri Lanka, limiting statistical generalizability beyond this multi-market context. The three-week data collection window and sample size of 103 respondents, while sufficient for the case study approach adopted, may not fully capture the breadth of perspectives across the globe. Additionally, the rapidly evolving nature of green finance regulation, illustrated by the US withdrawal from the Paris Agreement in early 2025 and the subsequent withdrawal of major US banks from the NZBA, means that certain policy findings may require reassessment as the global regulatory landscape continues to shift.

5.0 Conclusion

Stock exchanges play a transformative role in shaping the future of green finance. By creating platforms for the issuance of green financial instruments such as green bonds and fostering transparent reporting mechanisms, stock exchanges serve as pivotal enablers of sustainability. The insights offered by Grabski & Miller (2017) and Meng et al. (2017) reveal the substantial influence that exchanges can exert in catalyzing the adoption of green finance. The adoption of green finance practices by banks underscores the critical role of financial institutions in

fostering sustainable development. Banks in emerging markets are gradually embracing green finance innovations, yet challenges like insufficient standardization and transparency persist. Bergedieck et al. (2019) highlight the importance of enhancing collaboration between banks, regulators, and other stakeholders to overcome these hurdles. Stock exchanges influence banks by driving regulatory compliance, mandating sustainability disclosures, and creating incentives for the adoption of environmentally friendly practices. Regulations, mechanisms, awareness, and incentives collectively form the foundation for green finance adoption. Market regulations, as seen in the case of China and the EU (Larsen, 2022), provide a structured framework to align policies globally. Market mechanisms leverage technological advancements, as shown by Anbumozhi & Kimura (2018), to make green finance more efficient and accessible. Awareness campaigns, emphasized by Tröger and Steuer (2021), empower stakeholders with critical information, while incentives, such as tax credits and subsidies (Patel & Jose, 2024), make green finance more appealing. Together, these drivers ensure a holistic and sustainable approach to green finance. Ultimately, these findings carry direct relevance for policymakers, regulators, and exchange operators across emerging economies seeking evidence-based frameworks for accelerating the transition to sustainable finance.

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Declaration of generative AI use: The author(s) report generative AI was not used in their research or preparation of this manuscript.

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